1 Chapter 20 Crustaceans

1.1 Phylum Arthropoda: Subphylum Crustacea

Crustaceans are often called “Insects of the Sea” due to their dominance in marine habitats. They include lobsters, crayfishes, shrimps, crabs, and many smaller forms like copepods and water fleas.

1.1.1 General Characteristics

1. Appendages:
   - Two pairs of antennae (distinguishing characteristic from other arthropods).
   - Mandibles for chewing.
   - Two pairs of maxillae on the head.
   - One pair of appendages on each body segment (usually).
   - Ancestrally biramous (two main branches).
2. Respiration: Gills (in larger forms) or across the body surface (in smaller forms).
3. Tagmata: Usually Head, Thorax, and Abdomen.
   - Cephalothorax: Formed by the fusion of one or more thoracic segments with the head.
   - Carapace: Dorsal cuticle of the head that may extend posteriorly to cover the cephalothorax.

Figure 1: Cladogram showing hypothetical ancestor-descendant relationships of hexapods and classes of crustaceans. Hexapods and crustaceans are hypothetical sister groups evolving from a common ancestor defined by numerous shared derived characteristics. Characters followed by a question mark may be ancestral rather than shared derived features. The acron is the anterior region of the head and is not counted as a segment.

1.1.2 Class Malacostraca Body Plan

Malacostraca is the largest class (lobsters, crabs, etc.).

- Head: 5 fused segments.
- Thorax: 8 segments.
- Abdomen: 6 segments.
- Telson: Non-segmented posterior end.
- Rostrum: Non-segmented anterior end.

Figure 2: Archetypical plan of Malacostraca. The two maxillae and three maxillipeds have been separated diagrammatically to illustrate the general plan.

1.2 Form and Function (Model: Crayfish)

1.2.1 External Features

- Cuticle: Composed of chitin, protein, and calcareous material (harder in large crustaceans).
- Tergum: Dorsal cuticular plate.
- Sternum: Ventral transverse bar.
- Appendages: Exhibit serial homology (evolved from a common biramous ancestral form).
  - Protopod: Basal portion (Coxa + Basis).
  - Exopod: Lateral branch.
  - Endopod: Medial branch.

Figure 3: External structure of crayfishes. A, Dorsal view. B, Ventral view.

Figure 4: Parts of a biramous crustacean appendage (third maxilliped of a crayfish). The two branches of the appendage are the exopod and the endopod; both extend from the protopod.

Figure 5: Appendages of a crayfish showing variations on the basic biramous plan, as found in a swimmeret. Protopod, brown; endopod, blue; exopod, yellow.

Figure 6: Crayfish Appendages. Shows how various appendages have become modified from the presumed ancestral biramous plan and now perform disparate functions.

1.2.2 Internal Features

1. Muscular System: Striated muscles; antagonistic groups (flexors and extensors).
   - Abdominal flexors allow backward swimming (escape response).

Figure 7: Internal structure of a male crayfish.

2. Respiratory System:
   - Gills: Delicate, featherlike projections.
   - “Bailer” (part of 2nd maxilla) draws water over gill filaments.
3. Circulatory System:
   - Open system (lacunar).
   - Hemocoel: Major body space (persistent blastocoel).
   - Heart: Dorsal, single-chambered sac with ostia (valves).

- Pigments: Hemocyanin (copper, blue) or Hemoglobin (iron, red).

Figure 8: Diagrammatical cross section through heart region of a crayfish showing direction of blood flow in this “open” blood system. Heart pumps blood to body tissues through arteries, which empty into tissue sinuses. Returning blood enters sternal sinus, then goes through gills for gas exchange, and finally back to pericardial sinus by efferent channels. Note absence of veins.

4. Excretory System:
   - Antennal Glands (Green Glands) in decapods.
   - Structure: End sac (saccule + labyrinth) → Excretory tubule → Bladder → Pore.
   - Function: Regulation of ionic and osmotic composition (flood control in freshwater).
   - Nitrogenous waste (ammonia) diffuses chiefly across gills.

Figure 9: Scheme of antennal gland (green gland) of crayfishes. (In natural position organ is much folded.) Some crustaceans lack a labyrinth, and the excretory tubule (nephridial canal) is a much-coiled tube.

5. Nervous and Sensory Systems:
   - Supraesophageal ganglia (Brain).

- Subesophageal ganglion.
- Statocysts: At base of first antenna; contain statoliths (sand grains) for balance.
- Compound Eyes: Composed of ommatidia.
  - **Apposition image** (Mosaic): Strong light; pigment isolates ommatidia.
  - **Superposition image** (Continuous): Dim light; pigment retracts.

Figure 10: Portion of compound eye of an arthropod showing migration of pigment in ommatidia for day and night vision. Five ommatidia are represented in each diagram. In daytime each ommatidium is surrounded by a dark pigment collar so that each ommatidium is stimulated only by light rays that enter its own cornea (mosaic vision); in nighttime, pigment forms incomplete collars and light rays can spread to adjacent ommatidia (continuous, or superposition, image).

6. Reproduction:
   - Most dioecious (separate sexes).
   - Brooding of eggs is common.
   - Larva: Ancestral larva is the Nauplius (3 pairs of appendages: antennules, antennae, mandibles).
   - Metamorphosis: Change from larva to adult.

Figure 11: Life cycle of a Gulf shrimp, **Farfantepenaeus**. Penaeids spawn at depths of 40 to 90 m. Young larval forms are planktonic and move inshore to water of lower salinity to develop as benthic juveniles and adults. Older shrimp return to deeper water offshore.

7. Molting (Ecdysis):
   - Necessary for growth.
   - Process:
     1. Premolt: Epidermis separates, secretes new epicuticle.
     2. Enzymes dissolve old endocuticle; products resorbed.
     3. Ecdysis: Old cuticle sheds.
     4. Postmolt: Stretching and hardening (sclerotization/calcification).
   - Hormonal Control:
     - X-organ (Eyestalk): Produces Molt-inhibiting hormone.
     - Y-organ (near mandibles): Produces Molting hormone (ecdysone) when inhibition drops.
     - Androgenic glands: Stimulate male sexual characteristics.

Figure 12: Cuticle secretion and resorption in ecdysis.

Figure 13: Molting sequence in a lobster, **Homarus americanus**. A, Membrane between carapace and abdomen ruptures, and carapace begins slow elevation. This step may take up to 2 hours. B and C, Head, thorax, and finally abdomen withdraw. This process usually takes no more than 15 minutes. Immediately after ecdysis, chelipeds are desiccated and body is very soft. Lobster continues rapid absorption of water so that within 12 hours the body increases about 20% in length and 50% in weight. Tissue water will be replaced by protein in succeeding weeks.

8. Feeding:
   - Gastric Mill: Two-part stomach with calcareous teeth for grinding.

Figure 14: Malacostracan stomach showing gastric “mill” and directions of food movements. Mill has chitinous ridges, or teeth, for mastication, and setae for straining food before it passes into the pyloric stomach.

1.3 Classification of Crustaceans

1.3.1 Class Remipedia

- Ancestral features; cave-dwelling.
- 25-38 trunk segments with similar, paired, biramous swimming appendages.

Figure 15: A, A remipede crustacean of class Remipedia. B, A cephalocarid crustacean of class Cephalocarida.

1.3.2 Class Cephalocarida

- Tiny, bottom-dwelling.
- True hermaphrodites.

1.3.3 Class Branchiopoda

- Phyllopodia: Flattened, leaf-like legs for respiration and feeding.
- Orders:
  - Anostraca: Fairy shrimp, brine shrimp (no carapace).
  - Notostraca: Tadpole shrimp (dorsal shield).
  - Diplostraca: Water fleas (**Daphnia**). Important zooplankton.

Figure 16: Animals in A, B, and C are members of class Branchiopoda. A, Tadpole shrimp (order Notostraca). B, Fairy shrimp (order Anostraca). C, **Daphnia** (order Diplostraca, suborder Cladocera).

1.3.4 Class Ostracoda

- Enclosed in a bivalve carapace (resemble tiny clams).
- Fusion of trunk segments.

Figure 17: A, An ostracod of class Ostracoda. B, A mystacocarid crustacean of subclass Mystacocarida, class Maxillopoda. C, A copepod with attached ovisacs; subclass Copepoda, class Maxillopoda.

1.3.5 Class Maxillopoda

- Generally 5 cephalic, 6 thoracic, 4 abdominal segments.
- Naupliar eye (Maxillopodan eye).

1. Subclass Copepoda:
   - No carapace.
   - Antennules used for swimming.
   - Ecologically critical (primary consumers, food for fish/whales).
   - Many parasitic forms.
2. Subclass Branchiura (Fish Lice):
   - Ectoparasites on fish.
   - Suction cups (modified 2nd maxillae).

Figure 18: Fish louse; subclass Branchiura, class Maxillopoda.

3. Subclass Pentastomida (Tongue Worms):
   - Parasites of vertebrate respiratory systems.
   - Wormlike, segmented appearance (annuli).
   - 4 hooks + mouth.

Figure 19: Two pentastomids. A, **Linguatula**, found in nasal passages of carnivorous mammals. Female is shown with some internal structures. B, Female **Armillifer**, a pentastomid with pronounced body rings. In parts of Africa and Asia, humans are parasitized by immature stages; adults (10 cm long or more) live in lungs of snakes. Human infection may occur from eating snakes or from contaminated food or water.

Figure 20: Anterior end of a pentastome. Note both the mouth (arrow), between the middle hooks, and the apical sensory papillae.

4. Subclass Cirripedia (Barnacles):
   - Sessile adults (attached to substrate).
   - Carapace secretes calcareous plates.
   - Cirri: Feathery legs for filter feeding.
   - Rhizocephala (e.g., **Sacculina**): Parasitic barnacles on crabs; root-like system.

Figure 21: Barnacles; order Thoracica, subclass Cirripedia, class Maxillopoda. A, Acorn barnacles, **Balanus balanoides**, on an intertidal rock await the return of the tide. B, Common gooseneck barnacles, **Lepas anatifera**. Note the feeding legs, or cirri, on **Lepas**. Barnacles attach themselves to a variety of firm substrates, including rocks, pilings, and boat bottoms.

Figure 22: Life cycle of **Sacculina** (order Rhizocephala, subclass Cirripedia, class Maxillopoda), parasite of crabs (**Carcinus**).

Figure 23: A tantulocarid. This curious little parasite is shown attached to the first antenna of its copepod host at left; subclass Tantulocarida, class Maxillopoda.

1.3.6 Class Malacostraca

- Largest class.

1. Order Isopoda:
   - Dorsoventrally flattened.
   - No carapace.
   - Sessile eyes.
   - Includes terrestrial sow bugs/pill bugs.

Figure 24: A, Four pill bugs, **Armadillidium vulgare** (order Isopoda, class Malacostraca), common terrestrial forms. B, Freshwater sow bug, **Caecidotea** sp., an aquatic isopod.

Figure 25: An isopod parasite (**Anilocra** sp.) on a coney (**Cephalopholis fulvus**) inhabiting a Caribbean coral reef (order Isopoda, class Malacostraca).

2. Order Amphipoda:
   - Laterally compressed.
   - No carapace.
   - Sessile eyes.
   - Beach hoppers, **Gammarus**.

Figure 26: Marine amphipods. A, Free-swimming amphipod, **Anisogammarus** sp. B, Skeleton shrimp, **Caprella** sp., shown on a bryozoan colony, resemble praying mantids. C, **Phronima**, a marine pelagic amphipod, takes over the tunic of a salp (subphylum Urochordata, see Chapter 23). Swimming by means of its abdominal swimmerets, which protrude from the opening of the barrel-shaped tunic, the amphipod maneuvers to catch its prey. The tunic is not seen (order Amphipoda, class Malacostraca).

Figure 27: A, Head and mouth of a healthy California grey whale, **Eschrichtius robustus**, bearing its characteristic heavy load of barnacles (order Thoracica, subclass Cirripedia, class Maxillopoda) and cyamid parasites (order Amphipoda, class Malacostraca) (arrows). Note yellowish plates of baleen in mouth (p. 640). B, Cyamid parasites of grey whale. Unlike most amphipods, these are dorsoventrally flattened. They have sharp, grasping claws on their legs.

3. Order Euphausiacea (Krill):
   - Carapace fused with all thoracic segments but gills exposed.
   - Major food for baleen whales.

Figure 28: **Meganyctiphanes** (order Euphausiacea, class Malacostraca) “northern krill.”

4. Order Decapoda:
    - 5 pairs of walking legs (first pair often chelipeds with chelae/pincers).
    - 3 pairs of maxillipeds.
    - Crabs, lobsters, crayfish, shrimp.

Figure 29: Decapod crustaceans. A, A bright orange tropical rock crab, **Grapsus grapsus**, is a conspicuous exception to the rule that most crabs bear cryptic coloration. B, A hermit crab, **Elassochirus gilli**, which has a soft abdominal exoskeleton, lives in a snail shell that it carries about and into which it can withdraw for protection. C, A male fiddler crab, **Uca** sp., uses its enlarged cheliped to wave territorial displays and in threat and combat. D, A red night shrimp, **Rhynchocinetes rigens**, prowls caves and overhangs of coral reefs, but only at night. E, A spiny lobster, **Panulirus argus** (shown here), and the northern lobster, **Homarus americanus**, are consumed with gusto by many people (order Decapoda, class Malacostraca).

Figure 30: Sponge crab, **Dromidia antillensis**. This crab is one of several species that deliberately mask themselves with material from their immediate environment (order Decapoda, class Malacostraca).